

Abstract *EduAgent* is an educational AI agent that supports self-directed Python learning through guided tutoring, structured knowledge grounding, and visual learning feedback. Large introductory programming courses can enroll 700+ students, making timely individual support difficult; meanwhile, general-purpose AI tools may provide complete answers too readily and reduce active reasoning. To ground our design in learner needs, we conducted a formative study with 40 participants, developed *EduAgent* for Python novices, and evaluated it through a task-based pilot study with six undergraduates. Our long-term goal is scalable campus deployment for introductory Python education.

Formative Study: From Learner Needs to Design Goals To ground *EduAgent* in students’ actual learning needs, we surveyed 40 participants with programming and AI-use experience. We combined descriptive analysis with inductive thematic analysis to examine current AI use, perceived benefits and risks, and expectations for educational agents. As shown in the figure below, students valued AI’s efficiency and learning support but remained concerned about over-reliance, skill development, hallucinations, and reliability.

